lects; but it does not follow that the concept I have of myself is in any way different from the concept my *Doppelgänger* has of himself.

Now then, we have maintained that indexicality extends beyond the *obviously* indexical words and morphemes (e.g., the tenses of verbs). Our theory can be summarized as saying that words like "water" have an unnoticed indexical component: "water" is stuff that bears a certain similarity relation to the water *around here*. Water at another time or in another place or even in another possible world has to bear the relation same_L to our "water" *in order to be water*. Thus the theory that (1) words have "intensions," which are something like concepts associated with the words by speakers; and (2) intension determines extension — this theory cannot be true of natural-kind words like "water" for the same reason it cannot be true of obviously indexical words like "I."

The theory that natural-kind words like "water" are indexical leaves it open, however, whether to say that "water" in the Twin Earth dialect of English has the same *meaning* as "water" in the Earth dialect and a different extension (which is what we normally say about "I" in different idiolects), thereby giving up the doctrine that "meaning (intension) determines extension"; or to say, as we have chosen to do, that difference is extension is *ipso facto* a difference in meaning for natural-kind words, thereby giving up the doctrine that meanings are concepts, or, indeed, mental entities of *any* kind.

It should be clear, however, that Kripke's doctrine that natural-kind words are rigid designators and our doctrine that they are indexical are but two ways of making the same point. We heartily endorse what Kripke says when he writes:

> Let us suppose that we do fix the reference of a name by a description. Even if we do so, we do not then make the name synonymous with the description, but instead we use the name rigidly to refer to the object so named, even in talking about counterfactual situations where the thing named would not satisfy the description in question. Now, this is what I think is in fact true for those cases of naming where the reference is fixed by description. But, in fact, I also think, contrary to most recent theorists, that the reference of names is rarely or almost never fixed by means of description. And by this I do not just mean what Searle says: "It's not a single description, but rather a cluster, a family of properties that fixes the reference." I mean that properties in this sense are not used at all.[10]

[10] See Kripke's "Identity and Necessity," in M. Munitz, ed., *Identity and Individuation* (New York: New York University Press, 1972), p. 157.

Let's be realistic. I wish now to contrast my view with one which is popular, at least among students (it appears to arise spontaneously). For this discussion, let us take as our example of a natural-kind word the word *gold*. We will not distinguish between "gold" and the cognate words in Greek, Latin, etc. And we will focus on "gold" in the sense of gold in the solid state. With this understood, we maintain: "gold" has not changed its *extension* (or not changed it significantly) in two thousand years. Our methods of *identifying* gold have grown incredibly sophisticated. But the extension of χρυσὸς in Archimedes' dialect of Greek is the same as the extension of *gold* in my dialect of English.

It is possible (and let us suppose it to be the case) that just as there were pieces of metal which could not have been determined *not* to be gold prior to Archimedes, so there were or are pieces of metal which could not have been determined *not* to be gold in Archimedes' day, but which we can distinguish from gold quite easily with modern techniques. Let X be such a piece of metal. Clearly X does not lie in the extension of "gold" in standard English; my view is that it did not lie in the extension of χρυσὸς in Attic Greek, either, although an ancient Greek would have *mistaken* X for gold (or, rather, χρυσὸς).

The alternative view is that "gold" *means* whatever satisfies the *contemporary* "operational definition" of *gold*. "Gold" a hundred years ago meant whatever satisfied the "operational definition" of *gold* in use a hundred years ago; "gold" now means whatever satisfies the operational definition of *gold* in use in 1973; and χρυσὸς meant whatever satisfied the operational definition of χρυσὸς in use *then*.

One common motive for adopting this point of view is a certain skepticism about *truth*. On the view I am advocating, when Archimedes asserted that something was gold (χρυσὸς) he was not just saying that it had the superficial characteristics of gold (in exceptional cases, something may belong to a natural kind and *not* have the superficial characteristics of a member of that natural kind, in fact); he was saying that it had the same general *hidden structure* (the same "essence," so to speak) as any normal piece of local gold. Archimedes would have said that our hypothetical piece of metal X was gold, but he would have been *wrong*. But *who's to say* he would have been wrong?

The obvious answer is: *we are* (using the best theory available today). For most people either the question (*who's to say?*) has bite, and our

answer has no bite, or our answer has bite and the question has no bite. Why is this?

The reason, I believe, is that people tend either to be strongly antirealistic or strongly realistic in their intuitions. To a strongly antirealistic intuition it makes little sense to say that what is in the extension of Archimedes' term χρυσὸς is to be determined using *our* theory. For the antirealist does not see our theory and Archimedes' theory as two approximately correct descriptions of some fixed realm of theory-independent entities, and he tends to be skeptical about the idea of "convergence" in science — he does not think our theory is a *better* description of the *same* entities that Archimedes was describing. But if our theory is *just* our theory, then to use *it* in deciding whether or not X lies in the extension of χρυσὸς is just as arbitrary as using Neanderthal theory to decide whether or not X lies in the extension of χρυσὸς. The only theory that it is *not* arbitrary to use is the one the speaker himself subscribes to.

The trouble is that for a strong antirealist *truth* makes no sense except as an intra-theoretic notion.[11] The antirealist can use truth intra-theoretically in the sense of a "redundancy theory"; but he does not have the notions of truth and reference available *extra-theoretically*. But *extension is tied to the notion of truth*. The extension of a term is just what the term is *true of*. Rather than try to retain the notion of extension via an awkward operationalism, the antirealist should reject the notion of extension as he does the notion of truth (in any extra-theoretic sense). Like Dewey, for example, he can fall back on a notion of "warranted assertibility" instead of truth (relativized to the scientific method, if he thinks there is a *fixed* scientific method, or to the best methods available at the time, if he agrees with Dewey that the scientific method itself evolves). Then he can say that "X is gold (χρυσὸς)" was warrantedly assertible in Archimedes' time and is not warrantedly assertible today (indeed, this is a *minimal* claim, in the sense that it represents the minimum that the realist and the antirealist can agree on); but the assertion that X was in the extension of χρυσὸς will be rejected as meaningless, like the assertion that "X is gold (χρυσὸς)" was *true*.

It is well known that narrow operationalism cannot successfully ac-

[11] For a discussion of this point, see my "Explanation and Reference," in G. Pearce and P. Maynard, eds., *Conceptual Change* (Dordrecht: Reidel, 1973).

count for the actual use of scientific or common-sense terms. Loosened versions of operationalism, like Carnap's version of Ramsey's theory, agree with if they do not account for actual scientific use (mainly because the loosened versions agree with any possible use!), but at the expense of making the communicability of scientific results a *miracle*. It is beyond question that scientists use terms as if the associated criteria were not *necessary and sufficient conditions*, but rather *approximately* correct characterizations of some world of theory-independent entities, and that they talk as if later theories in a mature science were, in general, *better* descriptions of the *same* entities that earlier theories referred to. In my opinion the hypothesis that this is *right* is the only hypothesis that can account for the communicability of scientific results, the closure of acceptable scientific theories under first-order logic, and many other features of the scientific method.[12] But it is not my task to argue this here. My point is that if we are to use the notions of truth and extension in an extra-theoretic way (i.e., to regard those notions as defined for statements couched in the languages of theories other than our own), then we should accept the realist perspective to which those notions belong. The doubt about whether we can say that X does not lie in the extension of "gold" as *Jones* used it is the *same* doubt as the doubt whether it makes sense to think of Jones's statement that "X is gold" as *true* or *false* (and not just "warrantedly assertible for Jones and not warrantedly assertible for us"). To square the notion of truth, which is essentially a realist notion, with one's antirealist prejudices by adopting an untenable theory of meaning is no progress.

A second motive for adopting an extreme operationalist account is a dislike of unverifiable hypotheses. At first blush it may seem as if we are saying that "X is gold (χρυσὸς)" was false in Archimedes' time although Archimedes could not *in principle* have known that it was false. But this is not exactly the situation. The fact is that there are a host of situations that we can describe (using the very theory that tells us that X isn't gold) in which X would have behaved quite unlike the rest of the stuff Archimedes classified as gold. Perhaps X would have separated into two different metals when melted, or would have had different conductivity properties, or would have vaporized at a different tempera-

[12] For an illuminating discussion of just these points, see R. Boyd's "Realism and Scientific Epistemology" (unpublished; draft circulated by the author, Cornell University Department of Philosophy).

ture, or whatever. If we had performed the experiments with Archimedes watching, he might not have known the theory, but he would have been able to check the empirical regularity that "X behaves differently from the rest of the stuff I classify as χρυσὸς in several respects." Eventually he would have concluded that "X may not be gold."

The point is that even if something satisfies the criteria used at a given time to identify gold (i.e., to recognize if something is gold), it may behave differently in one or more situations from the rest of the stuff that satisfies the criteria. This may not *prove* that it isn't gold, but it puts the hypothesis that it may not be gold in the running, even in the absence of theory. If, now, we had gone on to inform Archimedes that gold has such and such a molecular structure (except for X), and that X behaved differently because it had a different molecular structure, is there any doubt that he would have agreed with us that X isn't gold? In any case, to worry because things may be *true* (at a given time) that can't be *verified* (at that time) seems to me ridiculous. On any reasonable view there are surely things that are true and can't be verified at *any* time. For example, suppose there are infinitely many binary stars. *Must* we be able to verify this, even *in principle*?[13]

So far we have dealt with *metaphysical* reasons for rejecting our account. But someone might disagree with us about the empirical facts concerning the intentions of speakers. This would be the case if, for instance, someone thought that Archimedes (in the *Gedankenexperiment* described above) would have said: "it doesn't matter if X *does* act differently from other pieces of gold; X is a piece of gold, because X has such-and-such properties and that's all it takes to be gold." While, indeed, we cannot be certain that natural-kind words in ancient Greek had the properties of the corresponding words in present-day English, there cannot be any serious doubt concerning the properties of the latter. If we put philosophical prejudices aside, then I believe that we know perfectly well that no operational definition does provide a necessary and sufficient condition for the application of any such word. We may

[13] See my "Logical Positivism and the Philosophy of Mind," in P. Achinstein, *The Legacy of Logical Positivism* (Baltimore: Johns Hopkins Press, 1969); and also my "Degree of Confirmation and Inductive Logic," in P. A. Schilpp, ed., *The Philosophy of Rudolf Carnap* (La Salle, Ill.: Open Court, 1962), and my "Probability and Confirmation" (broadcast for the Voice of America Philosophy of Science Series, Spring 1963; reprinted in A. Danto and S. Morgenbesser, eds., *Philosophy of Science Today* (New York: Basic Books, 1967).

give an "operational definition," or a cluster of properties, or whatever, but the intention is never to "make the name *synonymous* with the description." Rather "we use the name *rigidly*" to refer to whatever things share the *nature* that things satisfying the description normally possess.

Other senses. What we have analyzed so far is the predominant sense of natural-kind words (or, rather, the predominant *extension*). But natural-kind words typically possess a number of senses. (Ziff has even suggested that they possess a *continuum* of senses.)

Part of this can be explained on the basis of our theory. To be water, for example, is to bear the relation same_L to certain things. But what is the relation same_L?

x bears the relation same_L to y just in case (1) x and y are both liquids, and (2) x and y agree in important physical properties. The term "liquid" is itself a natural-kind term that I shall not try to analyze here. The term "property" is a broad-spectrum term that we have analyzed in previous papers. What I want to focus on now is the notion of *importance*. Importance is an interest-relative notion. Normally the "important" properties of a liquid or solid, etc., are the ones that are *structurally* important: the ones that specify what the liquid or solid, etc., is ultimately made out of — elementary particles, or hydrogen and oxygen, or earth, air, fire, water, or whatever — and how they are arranged or combined to produce the superficial characteristics. From this point of view the important characteristic of a typical bit of water is consisting of H_2O. But it may or may not be important that there are impurities; thus, in one context "water" may mean *chemically* pure water, while in another it may mean the stuff in Lake Michigan. And structure may sometimes be unimportant; thus one may sometimes refer to XYZ as water if one is *using* it as water. Again, normally it is important that water is in the liquid state; but sometimes it is unimportant, and one may refer to a single H_2O molecule as water, or to water vapor as water ("water in the air").

Even senses that are so far out that they have to be regarded as a bit "deviant" may bear a definite relation to the core sense. For example, I might say "did you see the lemon," meaning the *plastic* lemon. A less deviant case is this: we discover "tigers" on Mars. That is, they look just like tigers, but they have a silicon-based chemistry instead of a carbon-based chemistry. (A remarkable example of parallel evolution!)

Are Martian "tigers" tigers? It depends on the context.

In the case of this theory, as in the case of any theory that is orthogonal to the way people have thought about something previously, misunderstandings are certain to arise. One which has already arisen is the following: a critic has maintained that the *predominant* sense of, say, "lemon" is the one in which anything with (a sufficient number of) the superficial characteristics of a lemon is a lemon. The same critic has suggested that having the hidden structure — the genetic code — of a lemon is necessary to being a lemon only when "lemon" is used as a term of *science*. Both of these contentions seem to me to rest on a misunderstanding, or, perhaps, a pair of complementary misunderstandings.

The sense in which literally *anything* with the superficial characteristics of a lemon is necessarily a lemon, far from being the dominant one, is extremely deviant. In that sense something would be a lemon if it looked and tasted like a lemon, even if it had a silicon-based chemistry, for example, or even if an electron-microscope revealed it to be a *machine*. (Even if we include growing "like a lemon" in the superficial characteristics, this does not exclude the silicon lemon, if there are "lemon" trees on Mars. It doesn't even exclude the machine-lemon; maybe the tree is a machine too!

At the same time the sense in which to be a lemon something has to have the genetic code of a lemon is *not* the same as the technical sense (if there is one, which I doubt). The technical sense, I take it, would be one in which "lemon" was *synonymous* with a description which *specified* the genetic code. But when we said (to change the example) that to be *water* something has to be H_2O we did not mean, as we made clear, that the *speaker* has to *know* this. It is only by confusing *metaphysical* necessity with *epistemological* necessity that one can conclude that, if the (metaphysically necessary) truth-condition for being water is being H_2O, then "water" must be synonymous with H_2O — in which case it is certainly a term of science. And similarly, even though the predominant sense of "lemon" is one in which to be a lemon something has to have the genetic code of a lemon (I believe), it does not follow that "lemon" is synonymous with a description which specifies the genetic code explicitly or otherwise.

The mistake of thinking that there is an important sense of "lemon" (perhaps the predominant one) in which to have the superficial characteristics of a lemon is at least *sufficient* for being a lemon is more plausible if among the superficial characteristics one includes *being cross-fertile with lemons*. But the characteristic of being cross-fertile with lemons presupposes the notion of being a lemon. Thus, even if one can obtain a sufficient condition in *this* way, to take this as inconsistent with the characterization offered here is question-begging. Moreover, the characterization in terms of *lemon*-presupposing "superficial characteristics" (like being cross-fertile with *lemons*) gives no truth-condition which would enable us to decide which objects in other possible worlds (or which objects a million years ago, or which objects a million light years from here) are lemons. (In addition, I don't think this characterization, question-begging as it is, is *correct*, even as a sufficient condition. I think one could invent cases in which something which was not a lemon was cross-fertile with lemons and looked like a lemon, etc.)

Again, one might try to rule out the case of the machine-lemon (lemon-machine?) which "grows" on a machine-tree (tree-machine?) by saying that "growing" is not really *growing*. That is right; but it's right because *grow* is a natural-kind *verb*, and precisely the sort of account we have been presenting applies to *it*.

Another misunderstanding that should be avoided is the following: to take the account we have developed as implying that the members of the extension of a natural-kind word necessarily *have* a common hidden structure. It could have turned out that the bits of liquid we call "water" had *no* important common physical characteristics *except* the superficial ones. In that case the necessary and sufficient condition for being "water" would have been possession of sufficiently many of the superficial characteristics.

Incidentally, the last statement does not imply that water could have failed to have a hidden structure (or that water could have been anything but H_2O). When we say that it could have *turned out* that water had no hidden structure, what we mean is that a liquid with no hidden structure (i.e., many bits of different liquids, with nothing in common *except* superficial characteristics) could have looked like water, tasted like water, and have filled the lakes, etc., that are actually full of water. In short, we could have been in the same epistemological situation with respect to a liquid with no hidden structure as we were actually with

respect to water at one time. Compare Kripke on the "lectern made of ice."[14]

There are, in fact, almost continuously many cases. Some diseases, for example, have turned out to have no hidden structure (the only thing the paradigm cases have in common is a cluster of symptoms), while others have turned out to have a common hidden structure in the sense of an etiology (e.g., tuberculosis). Sometimes we still don't know; there is a controversy still raging about the case of multiple sclerosis.

An interesting case is the case of *jade*. Although the Chinese do not recognize a difference, the term "jade" applies to two minerals: jadeite and nephrite. Chemically, there is a marked difference. Jadeite is a combination of sodium and aluminum. Nephrite is made of calcium, magnesium, and iron. These two quite different microstructures produce the same unique textural qualities!

Coming back to the Twin Earth example, for a moment, if H_2O and XYZ had both been plentiful on Earth, then we would have had a case similar to the jadeite/nephrite case: it would have been correct to say that there were *two kinds of "water."* And instead of saying that "the stuff on Twin Earth turned out not to really be water," we would have to say "it turned out to be the XYZ *kind of water*."

To sum up: if there is a hidden structure, then generally it determines what it is to be a member of the natural kind, not only in the actual world, but in all possible worlds. Put another way, it determines what we can and cannot counterfactually suppose about the natural kind ("water could have all been vapor?" yes/"water could have been XYZ?" no). But the local water, or whatever, may have two or more hidden structures — or so many that "hidden structure" becomes irrelevant, and superficial characteristics become the decisive ones.

Other words. So far we have only used natural-kind words as examples, but the points we have made apply to many other kinds of words as well. They apply to the great majority of all nouns, and to other parts of speech as well.

Let us consider for a moment the names of artifacts — words like "pencil," "chair," "bottle," etc. The traditional view is that these words are certainly defined by conjunctions, or possibly clusters, of properties.

[14] See Kripke's "Identity and Necessity."

Anything with all of the properties in the conjunction (or sufficiently many of the properties in the cluster, on the cluster model) is necessarily a *pencil, chair, bottle,* or whatever. In addition, some of the properties in the cluster (on the cluster model) are usually held to be *necessary* (on the conjunction-of-properties model, *all* of the properties in the conjunction are necessary). *Being an artifact* is supposedly necessary, and belonging to a kind with a certain standard purpose — e.g., "pencils are artifacts," and "pencils are standardly intended to be written with" are supposed to be necessary. Finally, this sort of necessity is held to be *epistemic* necessity — in fact, analyticity.

Let us once again engage in science fiction. This time we use an example devised by Rogers Albritton. Imagine that we someday discover that *pencils are organisms.* We cut them open and examine them under the electron microscope, and we see the almost invisible tracery of nerves and other organs. We spy upon them, and we see them spawn, and we see the offspring grow into full-grown pencils. We discover that these organisms are not imitating other (artifactual) pencils — there are not and never were any pencils except these organisms. It is strange, to be sure, that there is *lettering* on many of these organisms — e.g., BONDED *Grants* DELUXE made in U.S.A. No. 2 — perhaps they are intelligent organisms, and this is their form of camouflage. (We also have to explain why no one ever attempted to manufacture pencils, etc., but this is clearly a possible world, in some sense).

If this is conceivable, and I agree with Albritton that it is, then it is epistemically possible that *pencils could turn out to be organisms.* It follows that *pencils are artifacts* is not epistemically necessary in the strongest sense and, a fortiori, not analytic.

Let us be careful, however. Have we shown that there is a possible world in which pencils are organisms? I think not. What we have shown is that there is a possible world in which certain organisms are the *epistemic counterparts* of pencils (the phrase is Kripke's). To return to the device of Twin Earth: imagine this time that pencils on Earth are just what we think they are, artifacts manufactured to be written with, while "pencils" on Twin Earth are organisms a la Albritton. Imagine, further, that this is totally unsuspected by the Twin Earthians — they have exactly the beliefs about "pencils" that we have about pencils. When we discovered this, we would not say: "some pencils are organisms." We would be far more likely to say: "the things on Twin Earth

that pass for pencils aren't really pencils. They're really a species of organism."

Suppose now the situation to be as in Albritton's example both on Earth and on Twin Earth. Then we would say "pencils are organisms." Thus, whether the "pencil-organisms" on Twin Earth (or in another possible universe) are really *pencils* or not is a function of whether or not the *local* pencils are organisms or not. If the local pencils are just what we think they are, then a possible world in which there are pencil-organisms is *not* a possible world in which *pencils are organisms*; there are *no* possible worlds in which pencils are organisms in this case (which is, of course, the actual one). That pencils are artifacts *is* necessary in the sense of true in all possible worlds — metaphysically necessary. But it doesn't follow that it's epistemically necessary.

It follows that "pencil" is not synonymous with any description — not even loosely synonymous with a *loose* description. When we use the word "pencil," we intend to refer to whatever has the same *nature* as the normal examples of the local pencils in the actual world. "Pencil" is just as *indexical* as "water" or "gold."

In a way, the case of pencils turning out to be organisms is complementary to the case we discussed some years ago[15] of cats turning out to be robots (remotely controlled from Mars). In his contribution to the present volume, Katz argues that we misdescribed this case: that the case should rather be described as its *turning out that there are no cats in this world*. Katz admits that we might say "Cats have turned out not to be animals, but robots"; but he argues that this is a semantically deviant sentence which is glossed as "the things I am referring to as 'cats' have turned out not to be animals, but robots." Katz's theory is bad linguistics, however. First of all, the explanation of how it is we can say "Cats are robots" is simply an all-purpose explanation of how we can say *anything*. More important, Katz's theory predicts that "Cats are robots" is *deviant*, while "There are no cats in the world" is nondeviant, in fact standard, in the case described. Now then, I don't deny that there *is* a case in which "There are not (and never were) any cats in the world" would be standard: we might (speaking epistemically) discover that we have been suffering from a collective hallucination. ("Cats" are like pink elephants.) But in the case I described, "Cats

[15] See my "It Ain't Necessarily So," *Journal of Philosophy*, 59(1962):658–671.

have turned out to be robots remotely controlled from Mars" is surely nondeviant, and "There are no cats in the world" is highly deviant.

Incidentally, Katz's account is not only bad linguistics; it is also bad as a rational reconstruction. The reason we *don't* use "cat" as synonymous with a description is surely that we know enough about cats to know that they do have a hidden structure, and it is good scientific methodology to use the name to refer rigidly to the things that possess that hidden structure, and not to whatever happens to satisfy some description. Of course, if we *knew* the hidden structure we could frame a description in terms of *it*; but we don't at this point. In this sense the use of natural-kind words reflects an important fact about our relation to the world: we know that there are kinds of things with common hidden structure, but we don't yet have the knowledge to describe all those hidden structures.

Katz's view has more plausibility in the "pencil" case than in the "cat" case, however. We think we *know* a necessary and sufficient condition for being a *pencil*, albeit a vague one. So it is possible to make "pencil" synonymous with a loose description. We *might* say, in the case that "pencils turned out to be organisms" *either* "Pencils have turned out to be organisms" or "There are no pencils in the world" — i.e., we might use "pencil" either as a natural-kind word or as a "one-criterion" word.[16]

On the other hand, we might doubt that there are any true one-criterion words in natural language, apart from stipulative contexts. Couldn't it turn out that pediatricians aren't doctors but Martian spies? Answer "yes," and you have abandoned the synonymy of "pediatrician" and "doctor specializing in the care of children." It seems that there is a strong tendency for words which are introduced as "one-criterion" words to develop a "natural kind" sense, with all the concomitant rigidity and indexicality. In the case of artifact-names, this natural-kind sense seems to be the predominant one.

(There is a joke about a patient who is on the verge of being discharged from an insane asylum. The doctors have been questioning him for some time, and he has been giving perfectly sane responses. They

[16] The idea of a "one-criterion" word, and a theory of analyticity based on this notion, appears in my "The Analytic and the Synthetic," in H. Feigl and G. Maxwell, eds., *Minnesota Studies in the Philosophy of Science*, vol. 3 (Minneapolis: University of Minnesota Press, 1962).

decide to let him leave, and at the end of the interview one of the doctors inquires casually, "What do you want to be when you get out?" "A teakettle." The joke would not be intelligible if it were literally inconceivable that a person could be a teakettle.)

There are, however, words which retain an almost pure one-criterion character. These are words whose meaning derives from a transformation: *hunter = one who hunts.*

Not only does the account given here apply to most nouns, but it also applies to other parts of speech. Verbs like "grow," adjectives like "red," etc., all have indexical features. On the other hand, some syncategorematic words seem to have more of a one-criterion character. "Whole," for example, can be explained thus: *The army surrounded the town* could be true even if the A division did not take part. *The whole army surrounded the town* means every part of the army (of the relevant kind, e.g., the A Division) took part in the action signified by the verb.[17]

Meaning. Let us now see where we are with respect to the notion of meaning. We have now seen that the extension of a term is not fixed by a concept that the individual speaker has in his head, and this is true both because extension is, in general, determined *socially* — there is division of linguistic labor as much as of "real" labor — and because extension is, in part, determined *indexically*. The extension of our terms depends upon the actual nature of the particular things that serve as paradigms,[18] and this actual nature is not, in general, fully known to the speaker. Traditional semantic theory leaves out only two contributions to the determination of extension — the contribution of society and the contribution of the real world!

We saw at the outset that meaning cannot be identified with extension. Yet it cannot be identified with "intension" either, if intension is something like an individual speaker's *concept*. What are we to do?

There are two plausible routes that we might take. One route would be to retain the identification of meaning with concept and pay the price of giving up the idea that meaning determines extension. If we followed this route, we might say that "water" has the same *meaning*

[17] This example comes from an analysis by Anthony Kroch, in his doctoral dissertation, M.I.T. Department of Linguistics, 1974.

[18] I *don't* have in mind the Flewish notion of "paradigm" in which any paradigm of a *K* is *necessarily* a *K* (in reality).

on Earth and on Twin Earth, but a different *extension*. (Not just a different *local* extension but a different *global* extension. The XYZ on Twin Earth isn't in the extension of the tokens of "water" that I utter, but it is in the extension of the tokens of "water" that my *Doppelgänger* utters, and this isn't just because Twin Earth is far away from me, since molecules of H_2O are in the extension of the tokens of "water" that I utter no matter how far away from me they are in space and time. Also, what I can counterfactually suppose water to be is different from what my *Doppelgänger* can counterfactually suppose "water" to be.) While this is the correct route to take for an *absolutely* indexical word like "I," it seems incorrect for the words we have been discussing. Consider "elm" and "beech," for example. If these are "switched" on Twin Earth, then surely we would *not* say that "elm" has the same meaning on Earth and Twin Earth, even if my *Doppelgänger's* stereotype of a beech (or an "elm," as he calls it) is identical with my stereotype of an elm. Rather, we would say that "elm" in my *Doppelgänger's* idiolect means *beech*. For this reason, it seems preferable to take a different route and identify "meaning" with an ordered pair (or possibly an ordered *n-tuple*) of entities, *one of which is the extension*. (The other components of the, so to speak, "meaning vector" will be specified later.) Doing this makes it trivially true that *meaning determines extension* (i.e., difference in extension is ipso facto difference in meaning), but totally abandons the idea that if there is a difference in the meaning my *Doppelgänger* and I assign to a word, then there *must* be some difference in our concepts (or in our psychological state). Following this route, we can say that my *Doppelgänger* and I *mean something different* when we say "elm," but this will not be an assertion about our psychological states. All this means is that the tokens of the word he utters have a different extension than the tokens of the word I utter; but this difference in extension is not a reflection of any difference in our individual linguistic competence considered in isolation.

If this is correct, and I think it is, then the traditional problem of meaning splits into two problems. The first problem is to account for the *determination of extension*. Since, in many cases, extension is determined socially and not individually, owing to the division of linguistic labor, I believe that this problem is properly a problem for sociolinguistics. Solving it would involve spelling out in detail exactly how the division of linguistic labor works. The so-called "causal theory of refer-

ence," introduced by Kripke for proper names and extended by us to natural-kind words and physical-magnitude terms,[19] falls into this province. For the fact that, in many contexts, we assign to the tokens of a name that I utter whatever referent we assign to the tokens of the same name uttered by the person from whom I acquired the name (so that the reference is transmitted from speaker to speaker, starting from the speakers who were present at the "naming ceremony," even though no fixed *description* is transmitted) is simply a special case of social cooperation in the determination of reference.

The other problem is to describe *individual competence*. Extension may be determined socially, in many cases, but we don't assign the standard extension to the tokens of a word W uttered by Jones *no matter how* Jones uses W. Jones has to have some particular ideas and skills in connection with W in order to play his part in the linguistic division of labor. Once we give up the idea that individual competence has to be so strong as to actually determine extension, we can begin to study it in a fresh frame of mind.

In this connection it is instructive to observe that nouns like "tiger" or "water" are very different from proper names. One can use the proper name "Sanders" correctly without knowing anything about the referent except that he is called "Sanders" — and even that may not be correct. ("Once upon a time, a very long time ago now, about last Friday, Winnie-the-Pooh lived in a forest all by himself under the name of Sanders.") But one cannot use the word tiger correctly, save *per accidens*, without knowing a good deal about tigers, or at least about a certain conception of tigers. In this sense concepts *do* have a lot to do with meaning.

Just as the study of the first problem is properly a topic in sociolinguistics, so the study of the second problem is properly a topic in psycholinguistics. To this topic we now turn.

Stereotypes and communication. Suppose a speaker knows that "tiger" has a set of physical objects as its extension, but no more. If he possesses normal linguistic competence in other respects, then he could use "tiger" in *some* sentences: for example, "tigers have mass," "tigers take up space," "give me a tiger," "is that a tiger?" etc. Moreover, the *socially determined* extension of "tiger" in these sentences would be

[19] In my "Explanation and Reference," in Pearce and Maynard, eds., *Conceptual Change*.

the standard one, i.e., the set of tigers. Yet we would not count such a speaker as "knowing the meaning" of the word *tiger*. Why not?

Before attempting to answer this question, let us reformulate it a bit. We shall speak of someone as having *acquired* the word "tiger" if he is able to use it in such a way that (1) his use passes muster (i.e., people don't say of him such things as "he doesn't know what a tiger *is*," "he doesn't know the meaning of the word 'tiger,'" etc.); and (2) his total way of being situated in the world and in his linguistic community is such that the socially determined extension of the word "tiger" in his idiolect is the set of tigers. Clause (1) means, roughly, that speakers like the one hypothesized in the preceding paragraph don't count as having acquired the word "tiger" (or whichever). We might speak of them, in some cases, as having *partially acquired* the word; but let us defer this for the moment. Clause (2) means that speakers on Twin Earth who have the same linguistic habits as we do, count as having acquired the word "tiger" only if the extension of "tiger" in their idiolect is the set of tigers. The burden of the preceding sections of this paper is that it does *not* follow that the extension of "tiger" in Twin Earth dialect (or idiolects) is the set of tigers merely because their linguistic habits are the same as ours; the nature of Twin Earth "tigers" is also relevant. (If Twin Earth organisms have a silicon chemistry, for example, then their "tigers" aren't really tigers, even if they look like tigers, although the linguistic habits of the lay Twin Earth speaker exactly correspond to those of Earth speakers.) Thus clause (2) means that in this case we have decided to say that Twin Earth speakers have not acquired our word "tiger" (although they have acquired another word with the same spelling and pronunciation).

Our reason for introducing this way of speaking is that the question "does he know the meaning of the word 'tiger'?" is biased in favor of the theory that acquiring a word is coming to possess a thing called its "meaning." Identify this thing with a concept, and we are back at the theory that a sufficient condition for acquiring a word is associating it with the right concept (or, more generally, being in the right psychological state with respect to it) — the very theory we have spent all this time refuting. So, henceforth, we will "acquire" words, rather than "learn their meaning."

We can now reformulate the question with which this section began. The use of the speaker we described does not pass muster, although it

is not such as to cause us to assign a nonstandard extension to the word "tiger" in his idiolect. Why doesn't it pass muster?

Suppose our hypothetical speaker points to a snowball and asks, "is that a tiger?" Clearly there isn't much point in talking tigers with *him*. Significant communication requires that people know something of what they are talking about. To be sure, we hear people "communicating" every day who clearly know nothing of what they are talking about; but the sense in which the man who points to a snowball and asks "is that a tiger" doesn't know anything about tigers is so far beyond the sense in which the man who thinks that Vancouver is going to win the Stanley Cup, or that the Vietnam War was fought to help the South Vietnamese, doesn't know what he is talking about as to boggle the mind. The problem of people who think that Vancouver is going to win the Stanley Cup, or that the Vietnam War was fought to help the South Vietnamese, is one that obviously cannot be remedied by the adoption of linguistic conventions; but not knowing what one is talking about in the second, mind-boggling sense can be and is prevented, near enough, by our conventions of language. What I contend is that speakers are *required* to know something about (stereotypical) tigers in order to count as having acquired the word "tiger"; something about elm trees (or, anyway, about the stereotype thereof) to count as having acquired the word "elm"; etc.

This idea should not seem too surprising. After all, we do not permit people to drive on the highways without first passing some tests to determine that they have a *minimum* level of competence; and we do not dine with people who have not learned to use a knife and fork. The linguistic community too has its minimum standards, with respect both to syntax and to "semantics."

The nature of the required minimum level of competence depends heavily upon both the culture and the topic, however. In our culture speakers are required to know what tigers look like (if they acquire the word "tiger," and this is virtually obligatory); they are not required to know the fine details (such as leaf shape) of what an elm tree looks like. English speakers are *required by their linguistic community* to be able to tell tigers from leopards; they are not required to be able to tell elm trees from beech trees.

This could easily have been different. Imagine an Indian tribe, call it the Cheroquoi, who have words, say *uhaba'* and *wa'arabi* for elm trees

and beech trees respectively, and who make it *obligatory* to know the difference. A Cheroquoi who could not recognize an elm would be said not to know what an uhaba' is, not to know the meaning of the word "uhaba'" (perhaps, not to know the word, or not to *have* the word), just as an English speaker who had no idea that tigers are striped would be said not to know what a tiger is, not to know the meaning of the word "tiger" (of course, if he at least knows that tigers are large felines we might say he knows part of the meaning, or partially knows the meaning), etc. Then the translation of "uhaba'" as "elm" and "wa'-arabi" as "beech" would, on our view, be only *approximately* correct. In this sense there is a real difficulty with radical translation,[20] but this is not the abstract difficulty that Quine is talking about.[21]

What stereotypes are. I introduced the notion of a "stereotype" in my lectures at the University of Washington and at the Minnesota Center for the Philosophy of Science in 1968. I will not review all the argumentation from the subsequently published "Is Semantics Possible" in the present essay, but I do want to introduce the notion again and to answer some questions that have been asked about it.

In ordinary parlance a "stereotype" is a conventional (frequently malicious) idea (which may be wildly inaccurate) of what an X looks like or acts like or is. Obviously, I am trading on some features of the ordinary parlance. I am not concerned with malicious stereotypes (save where the language itself is malicious); but I am concerned with conventional ideas, which may be inaccurate. I am suggesting that just such a conventional idea is associated with "tiger," with "gold," etc., and, moreover, that this is the sole element of truth in the "concept" theory.

On this view someone who knows what "tiger" means (or, as we have decided to say instead, has acquired the word "tiger") is *required* to know that *stereotypical* tigers are striped. More precisely, there is *one* stereotype of tigers (he may have others) which is required by the linguistic community as such; he is required to have this stereotype, and to know (implicitly) that it is obligatory. This stereotype must in-

[20] The term is due to Quine (in *Word and Object*); it signifies translation without clues from either shared culture or cognates.

[21] For a discussion of the supposed impossibility of uniquely correct radical translation, see my "The Refutation of Conventionalism" (forthcoming in *Noûs* and also, in a longer version, in a collection edited by M. Munitz to be published by New York University Press under the title *Semantics and Philosophy*).

clude the feature of stripes if his acquisition is to count as successful.

The fact that a feature (e.g. stripes) is included in the stereotype associated with a word X does not mean that it is an analytic truth that all Xs have that feature, nor that most Xs have that feature, nor that all normal Xs have that feature, nor that some Xs have that feature.[22] Three-legged tigers and albino tigers are not logically contradictory entities. Discovering that our stereotype has been based on nonnormal or unrepresentative members of a natural kind is not discovering a logical contradiction. If tigers lost their stripes they would not thereby cease to be tigers, nor would butterflies necessarily cease to be butterflies if they lost their wings.

(Strictly speaking, the situation is more complicated than this. It is possible to give a word like "butterfly" a sense in which butterflies would cease to be butterflies if they lost their wings — through mutation, say. Thus one can find a sense of "butterfly" in which it is analytic that "butterflies have wings." But the most important sense of the term, I believe, is the one in which the wingless butterflies would still be butterflies.)

At this point the reader may wonder what the value to the linguistic community of having stereotypes is, if the "information" contained in the stereotype is not necessarily correct. But this is not really such a mystery. Most stereotypes do in fact capture features possessed by paradigmatic members of the class in question. Even where stereotypes go wrong, the way in which they go wrong sheds light on the contribution normally made by stereotypes to communication. The stereotype of gold, for example, contains the feature *yellow* even though chemically pure gold is nearly white. But the gold we see in jewelry is typically yellow (due to the presence of copper), so the presence of this feature in the stereotype is even useful in lay contexts. The stereotype associated with *witch* is more seriously wrong, at least if taken with existential import. Believing (with existential import) that witches enter into pacts with Satan, that they cause sickness and death, etc., facilitates communication only in the sense of facilitating communication internal to witch-theory. It does not facilitate communication in any situation in which what is needed is more agreement with the world than agreement with the theory of other speakers. (Strictly speaking, I am speaking of

[22] This is argued in "Is Semantics Possible?"

the stereotype as it existed in New England three hundred years ago; today that witches aren't *real* is itself part of the stereotype, and the baneful effects of witch-theory are thereby neutralized.) But the fact that our language has *some* stereotypes which impede rather than facilitate our dealings with the world and each other only points to the fact that we aren't infallible beings, and how could we be? The fact is that we could hardly communicate successfully if most of our stereotypes weren't pretty accurate as far as they go.

The "operational meaning" of stereotypes. A trickier question is this: how far is the notion of stereotype "operationally definable." Here it is necessary to be extremely careful. Attempts in the physical sciences to *literally* specify operational definitions for terms have notoriously failed; and there is no reason the attempt should succeed in linguistics when it failed in physics. Sometimes Quine's arguments against the possibility of a theory of meaning seem to reduce to the demand for operational definitions in linguistics; when this is the case the arguments should be ignored. But it frequently happens that terms do have operational definitions not in the actual world but in idealized circumstances. Giving these "operational definitions" has heuristic value, as idealization frequently does. It is only when we mistake operational definition for more than convenient idealization that it becomes harmful. Thus we may ask what is the "operational meaning" of the statement that a word has such and such a stereotype, without supposing that the answer to this question counts as a theoretical account of what it is to be a stereotype.

The theoretical account of what it is to be a stereotype proceeds in terms of the notion of *linguistic obligation*; a notion which we believe to be fundamental to linguistics and which we shall not attempt to explicate here. What it means to say that being striped is part of the (linguistic) stereotype of "tiger" is that it is *obligatory* to acquire the information that stereotypical tigers are striped if one acquires the word "tiger," in the same sense of "obligatory" as that it is obligatory to indicate whether one is speaking of lions in the singular or lions in the plural when one speaks of lions in English. To describe an idealized experimental test of this hypothesis is not difficult. Let us introduce a person whom we may call the linguist's *confederate*. The confederate will be (or pretend to be) an adult whose command of English is generally excellent, but who for some reason (raised in an alien culture?

brought up in a monastery?) has totally failed to acquire the word "tiger." The confederate will say the word "tiger" or, better yet, point to it (as if he wasn't sure how to pronounce it), and ask "what does this word mean?" or "what is this?" or some such question. Ignoring all the things that go wrong with experiments in practice, what our hypothesis implies is that informants should typically tell the confederate that tigers are, inter alia, striped.

Instead of relying on confederates, one might expect the linguist to study children learning English. But children learning their native language aren't taught it nearly as much as philosophers suppose; they learn it but they aren't taught it, as Chomsky has emphasized. Still, children do sometimes ask such questions as "what is a tiger?" and our hypothesis implies that in these cases too informants should tell them, inter alia, that tigers are striped. But one problem is that the informants are likely to be parents, and there are the vagaries of parental time, temper, and attention to be allowed for.

It would be easy to specify a large number of additional "operational" implications of our hypothesis, but to do so would have no particular value. The fact is that we are fully competent speakers of English ourselves, with a devil of a good sense of what our linguistic obligations are. Pretending that we are in the position of Martians with respect to English is not the route to methodological clarity; it was, after all, only when the operational approach was abandoned that transformational linguistics blossomed into a handsome science.

Thus if anyone were to ask me for the meaning of "tiger," I know perfectly well what I would tell him. I would tell him that tigers were feline, something about their size, that they are yellow with black stripes, that they (sometimes) live in the jungle, and are fierce. Other things I might tell him too, depending on the context and his reason for asking; but the above items, save possibly for the bit about the jungle, I would regard it as *obligatory* to convey. I don't have to experiment to know that this is what I regard it as obligatory to convey, and I am sure that approximately this is what other speakers regard it as obligatory to convey too. Of course, there is some variation from idiolect to idiolect; the feature of having stripes (apart from figure-ground relations, e.g., are they black stripes on a yellow ground, which is the way I see them, or yellow stripes on a black ground?) would be found in all normal idiolects, but some speakers might regard the information that tigers (stereo-

typically) inhabit jungles as obligatory, while others might not. Alternatively, some features of the stereotype (big-cat-hood, stripes) might be regarded as obligatory, and others as *optional*, on the model of certain syntactical features. But we shall not pursue this possibility here.

Quine's "Two Dogmas" revisited. In "Two Dogmas of Empiricism" Quine launched a powerful and salutory attack on the currently fashionable analytic-synthetic distinction. The distinction had grown to be a veritable philosophical man-eater: analytic *equaling* necessary *equaling* unrevisable in principle *equaling* whatever truth the individual philosopher wished to explain away. But Quine's attack itself went too far in certain respects; some limited class of analytic sentences can be saved, we feel.[23] More importantly, the attack was later construed, both by Quine himself and by others, as implicating the whole notion of meaning in the downfall of the analytic-synthetic distinction. While we have made it clear that we agree that the traditional notion of meaning has serious troubles, our project in this paper is constructive, not destructive. We come to revise the notion of meaning, not to bury it. So it will be useful to see how Quine's arguments fare against our revision.

Quine's arguments against the notion of analyticity can basically be reduced to the following: that no behavioral significance can be attached to the notion. His argument (again simplifying somewhat) was that there are, basically, only two candidates for a behavioral index of analyticity, and both are totally unsatisfactory, although for different reasons. The first behavioral index is *centrality*: many contemporary philosophers call a sentence analytic if, in effect, some community (say, Oxford dons) holds it immune from revision. But, Quine persuasively argues, maximum immunity from revision is no exclusive prerogative of analytic sentences. Sentences expressing fundamental laws of physics (e.g. the conservation of energy) may well enjoy maximum behavioral immunity from revision, although it would hardly be customary or plausible to classify them as analytic. Quine does not, however, rely on the mere implausibility of classifying all statements that we are highly reluctant to give up as analytic; he points out that "immunity from revision" is, in the actual history of science, a *matter of degree*. There is no such thing, in the actual practice of rational science, as *absolute* immunity from revision. Thus to identify analyticity with immunity from revision would alter the notion in two fundamental ways: analyticity

[23] See "The Analytic and the Synthetic."